

## Exercise 2.

① SELECT DISTINCT \*  
FROM student

| Department |
|------------|
| IT         |
| HR         |
| Finance    |

② SELECT \*  
FROM student  
GROUP BY department  
HAVING AVG(age) > 20

| Department | avg - age |
|------------|-----------|
| IT         | 20.5      |
| HR         | 22        |
| Finance    | 23        |

③ SELECT \*  
FROM student  
GROUP BY department  
HAVING COUNT(student id) > 1

| department | student - count |
|------------|-----------------|
| IT         | 2               |
| HR         | 2               |

④ SELECT \*  
FROM student  
WHERE age BETWEEN 21 AND 23

| student id | name    | age | department |
|------------|---------|-----|------------|
| 2          | Bob     | 22  | HR         |
| 3          | Charlie | 21  | IT         |
| 4          | Diana   | 23  | Finance    |
| 5          | Eve     | 22  | HR         |

⑤ SELECT \*  
FROM student  
WHERE department IN ('IT', 'HR')  
AND age > 21

| student id | name | age | department |
|------------|------|-----|------------|
| 2          | Bob  | 22  | HR         |
| 5          | Eve  | 22  | HR         |

⑥ SELECT \*  
FROM courses  
GROUP BY department  
HAVING SUM(credits) > 5

| department | total credits |
|------------|---------------|
| IT         | 10            |



7) SELECT \* course-id, course-name, department, credits  
FROM \* courses  
WHERE \* credits != 4;

| course id | course name | department | credits |
|-----------|-------------|------------|---------|
| 101       | SQL Basics  | IT         | 3       |
| 104       | Excel       | Finance    | 2       |
| 105       | Statistics  | HR         | 3       |

8) SELECT \* course-id, course-name, credits  
FROM \* courses  
ORDER BY \* credits DESC  
LIMIT 3;

| course id | course name  | credits |
|-----------|--------------|---------|
| 102       | Python       | 4       |
| 103       | Data Science | 4       |
| 101       | SQL Basics   | 3       |

9) SELECT  
MAX(grade) AS max-grade  
MIN(grade) AS min-grade  
AVG(grade) AS avg-grade  
FROM enrollments;

| Max-grade | min-grade | avg-grade |
|-----------|-----------|-----------|
| 90        | 75        | 84.6      |

10) SELECT \* course-id  
COUNT(enrollment-id) AS enrollment-count  
FROM enrollment  
GROUP BY course-id;

| course id | enrollment - count |
|-----------|--------------------|
| 101       | 2                  |
| 102       | 1                  |
| 103       | 1                  |
| 104       | 1                  |
| 105       | 1                  |